

TWO-COLLISION COUNTING RESPONSE IN A PERIODIC LORENTZ GAS

QIAN QI

ABSTRACT. For the periodic triangular Lorentz gas with disk radius in $[9/20, 47/100]$, we determine the full collision-count law over the physical time interval of length $11/100$. The window admits two collisions with positive probability for every radius, rather than excluding the second collision. Stationary flux identifies the two-collision probability with a positive-part integral of the genuine free-flight roof. A geometric short-flight estimate makes the entire support of this integral uniformly transverse, and its terminal level is a regular level throughout the radius interval. We prove smooth radius response of all three count probabilities, strict response of the two-collision probability, and explicit formulae for the moving-level terms at every finite order. No independence of consecutive collisions is imposed. We also give the exact collision-tail formula for arbitrary bounded windows. The results concern count statistics under the full equilibrium preparation; they do not assert a long-time susceptibility theorem or response of the entire two-collision path distribution.

1. THE MECHANICAL FAMILY AND THE RESULT

Let $e_1 = (1, 0)$, $e_2 = (1/2, \sqrt{3}/2)$, and $\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2$. On $\mathbb{T}_\Lambda^2 = \mathbb{R}^2/\Lambda$ remove the disk of radius R , where

$$I = [9/20, 47/100], \quad A_R = \sqrt{3}/2 - \pi R^2, \quad T = 11/100. \quad (1.1)$$

The particle has unit speed and undergoes Euclidean elastic reflection. Its preparation is the entire normalized phase volume

$$\mu_R(dq, d\vartheta) = \frac{\mathbf{1}_{Q_R}(q)}{2\pi A_R} dq d\vartheta, \quad Q_R = \mathbb{T}_\Lambda^2 \setminus \overline{B}(0, R).$$

Let N_T count collisions in $(0, T]$. Endpoint and tangent conventions alter no probabilities below.

On the outgoing collision section write $q = Rn_\alpha$ and

$$v = \cos \phi n_\alpha + \sin \phi t_\alpha, \quad n_\alpha = (\cos \alpha, \sin \alpha), \quad t_\alpha = (-\sin \alpha, \cos \alpha), \quad |\phi| < \pi/2.$$

The normalized section measure and its mean physical roof are

$$d\nu = \frac{\cos \phi}{4\pi} d\alpha d\phi, \quad \bar{\tau}_R = \int \tau_R d\nu = \frac{A_R}{2R}, \quad \lambda_R = \bar{\tau}_R^{-1}. \quad (1.2)$$

Here τ_R is the actual next collision time, not a prescribed symbolic clock. Put

$$J(R) = \int (T - \tau_R)_+ d\nu. \quad (1.3)$$

Theorem 1.1. *For the family (1.1), $N_T \in \{0, 1, 2\}$ almost surely and*

$$p_2 = \lambda_R J(R), \quad p_1 = \lambda_R T - 2\lambda_R J(R), \quad p_0 = 1 - \lambda_R T + \lambda_R J(R). \quad (1.4)$$

Date: September 8, 2026.

2020 Mathematics Subject Classification. 37D50, 37A50, 60G55.

Key words and phrases. Lorentz gas, collision counts, moving boundaries, suspension flow, response.

All three probabilities are strictly positive and extend smoothly to a neighborhood of I . Every fixed finite derivative is bounded uniformly on I . Moreover

$$p_2(R) \geq 10^{-9}, \quad p'_2(R) > 0 \quad (R \in I).$$

For each nearest lattice center e , let $\tau_{e,R}$ be the smooth entry root on the short-flight branch, and let $D_e(R) = \{\tau_{e,R} < T\}$ in section coordinates. Then

$$J'(R) = - \sum_e \int_{D_e(R)} \partial_R \tau_{e,R} d\nu, \quad (1.5)$$

$$J''(R) = - \sum_e \int_{D_e(R)} \partial_R^2 \tau_{e,R} d\nu + \sum_e \int_{\tau_{e,R}=T} \frac{(\partial_R \tau_{e,R})^2 \cos \phi}{4\pi |\nabla_{\alpha,\phi} \tau_{e,R}|} d\mathcal{H}^1. \quad (1.6)$$

The formulas include the preparation normalizer through (1.4). They do not require a grazing cutoff on the initial equilibrium preparation.

The integral (1.3) is a deterministic flux integral. We do not introduce a renewal assumption. The stationary counting identities used in the proof are elementary suspension identities, closely related to the Palm description of entry times [1]. The argument combines this stationary identity with regularity of the complete short-flight region. In particular, the one-event computation is not simply iterated.

2. FLUX AND EXACT COLLISION TAILS

2.1. Geometry and equilibrium suspension. The shortest nonzero lattice vector has length one. Distinct disks are separated by at least

$$d_* = 1 - 2 \max I = 3/50. \quad (2.1)$$

An outgoing ray cannot hit the same convex disk again. Thus successive collisions have gap at least d_* . There can be no finite-time collision accumulation. Grazing directions from a fixed exterior position form a finite or countable null set; regular collision maps preserve flux, and their finite preimages are null. Removing their countable union gives the full-measure regular dynamics used here.

For completeness the family has finite horizon. For a primitive lattice direction w , parallel lattice rows have spacing $(\sqrt{3}/2)/|w| \leq \sqrt{3}/2$. A parallel line has distance at most $\sqrt{3}/4 < 9/20$ from a row, and eventually passes through a disk. An irrational-direction line is dense on the torus. If arbitrarily long collision-free segments existed at the smallest radius, a convergent sequence of starting positions and directions would yield an infinite ray avoiding every open disk, a contradiction. Compactness gives a common finite roof upper bound. Enlarging the disks preserves this upper bound.

A boundary flow tube has volume $\cos \phi ds d\phi dt$, where $ds = R d\alpha$. Its total section flux is $4\pi R$, while total unit-speed phase volume is $2\pi A_R$. Consequently

$$2\pi A_R = 4\pi R \int \tau_R d\nu.$$

Reflection reverses normal velocity and preserves flux. Free flight preserves phase volume. The collision map B_R therefore preserves ν , and (1.2) follows. The flow preparation is the suspension measure

$$\frac{d\nu(x) ds}{\bar{\tau}_R}, \quad 0 \leq s < \tau_R(x). \quad (2.2)$$

No mixing property enters this statement.

2.2. A general counting identity. Let B be an invertible probability-preserving map with roof $\tau \geq d_* > 0$ and $\bar{\tau} = \int \tau d\nu < \infty$. Write $S_j\tau(x) = \sum_{h=0}^{j-1} \tau(B^h x)$, with $S_0 = 0$.

Proposition 2.1. *Under the stationary suspension preparation, for every $k \geq 1$ and $t \geq 0$,*

$$\Pr(N_t \geq k) = \frac{1}{\bar{\tau}} \int [(t - S_{k-1}\tau)_+ - (t - S_k\tau)_+] d\nu. \quad (2.3)$$

In particular $\mathbb{E}N_t = t/\bar{\tau}$. These formulas hold without independent roofs or a Markov hypothesis.

Proof. From a suspension point (x, s) , the k th future collision occurs after $S_k\tau(x) - s$. The length of the subset of $[0, \tau(x))$ for which this is at most t is

$$(t - S_{k-1}\tau(Bx))_+ - (t - S_k\tau(x))_+.$$

This follows by intersecting the interval $s \geq S_k\tau(x) - t$ with $[0, \tau(x))$. Integrate using (2.2). Invariance of ν replaces Bx by x in the first term, proving (2.3). Because $S_j\tau \geq jd_*$, only finitely many tail terms are nonzero for fixed t . Summing in k telescopes to $t/\bar{\tau}$. $\square$

When $T < 2d_*$, the second tail is $\lambda_R J(R)$ and every third tail is zero. Also the first tail is $\lambda_R(T - J(R))$. Taking differences proves (1.4). Notice the correction to the former one-event identity:

$$\Pr(N_T \geq 1) = \lambda_R T - p_2, \quad \mathbb{E}N_T = \lambda_R T. \quad (2.4)$$

Once a second event is possible, event probability and expected count are different.

More generally, with $A_j(t) = \int (t - S_j\tau)_+ d\nu$, the probability of exactly $k \geq 1$ collisions is

$$\bar{\tau}^{-1}(A_{k-1} - 2A_k + A_{k+1}).$$

This is an exact decomposition at every bounded horizon. It does not supply the parameter regularity of the roof iterates $S_j\tau$.

3. THE COMPLETE SHORT-FLIGHT REGION IS REGULAR

Let $q = Rn_\alpha$, $v = \cos \phi n_\alpha + \sin \phi t_\alpha$ and suppose the next impact is at $q' = e + Rn_1 = q + \tau v$. Set $c_0 = n_\alpha \cdot v > 0$ and $c_1 = -n_1 \cdot v > 0$.

Lemma 3.1. *Every next flight with $\tau \leq T$ ends at one of the six nearest lattice centers. Both endpoint cosines satisfy*

$$c_0, c_1 \geq \frac{1 - 2R - T^2}{2RT} \geq \frac{479}{1034} > \frac{46}{100}. \quad (3.1)$$

All entry roots with length at most T are therefore in finitely many smooth branch neighborhoods. No other disk obstructs such a root.

Proof. The triangle inequality gives $|e| \leq 2R + T \leq 1.05 < \sqrt{3}$. The next possible lattice length after one is $\sqrt{3}$, so $|e| = 1$. Since $|q'| \geq 1 - R$,

$$(1 - R)^2 \leq |q'|^2 = R^2 + 2R\tau c_0 + \tau^2.$$

The analogous identity $|q - e|^2 = R^2 + 2R\tau c_1 + \tau^2$ proves the same bound for c_1 . The function $(1 - 2R - \tau^2)/(2R\tau)$ decreases in both variables in these ranges. Its value at $R = 47/100$, $\tau = 11/100$ is $479/1034$. A hypothetical earlier different obstacle followed by the candidate obstacle along the straight segment would require at least two inter-obstacle gaps, exceeding T ; a repeated encounter with the initial disk is excluded by convexity. Thus every outgoing entry root of length at most T is the first actual hit. The implicit root denominator is $-2Rc_1$, uniformly separated from zero. Compactness and the six labels give the claimed neighborhoods. $\square$

This lemma does not say that all first collisions from arbitrary phase points are transverse. In the stationary count identity, the integral concerns the complete intercollision flight following a section point. Grazing first events remain in the equilibrium ensemble. The count statistic has eliminated their individual decoding through an exact identity, not by assigning them zero probability.

On a branch let $d = e - Rn_\alpha$, $L = d \cdot v$, and $b = d \cdot v^\perp$. Then

$$\tau_{e,R} = L - \sqrt{R^2 - b^2}, \quad \partial_R \tau_{e,R} = \frac{1 - n_1 \cdot n_\alpha}{n_1 \cdot v} < 0. \quad (3.2)$$

The root formula solves $|q + tv - e|^2 = R^2$ on its entry branch. Differentiating that equation at fixed (α, ϕ) gives the derivative formula; v is independent of R in these coordinates. If its numerator vanished, $n_1 = n_\alpha$, contrary to $n_\alpha \cdot v > 0 > n_1 \cdot v$. It is thus strictly positive.

Lemma 3.2. *The level $\tau_{e,R} = T$ is regular in the two section variables. The regularity is uniform for $R \in I$ over its complete level set.*

Proof. Use (α, ϑ) , where $\vartheta = \alpha + \phi$ is the direction angle. This is a fixed invertible linear change of angular coordinates. Implicit differentiation gives

$$\partial_\vartheta \tau = -\tau \frac{n_1 \cdot v^\perp}{n_1 \cdot v}, \quad \partial_\alpha \tau|_\vartheta = -R \frac{n_1 \cdot t_\alpha}{n_1 \cdot v}.$$

If both vanished, n_1 would be parallel to both v and n_α . The incoming and outgoing signs force $v = n_\alpha$ and $n_1 = -n_\alpha$. The chord equation becomes $e = (2R + \tau)n_\alpha$, so $\tau = 1 - 2R \leq 1/10 < T$. This contradicts the level equation. The short-flight support is compactly separated from tangency by Lemma 3.1; the level sets form a compact subset of the finitely many branch neighborhoods. Continuity gives a positive lower gradient bound there. All inequalities are strict on I , so the result extends to an open radius neighborhood. $\square$

4. MOVING-LEVEL RESPONSE AT EVERY FINITE ORDER

4.1. Smoothness of the positive-part integral. The complete active set $\{\tau_R \leq T\}$ is contained in the regular branch neighborhoods of Lemma 3.1. Choose a smooth partition of unity on a neighborhood of its compact union in parameter–section space. Terms away from $\tau = T$ are ordinary smooth parameter integrals: the positive part either vanishes or is $T - \tau$. Near the level, the parametric implicit function theorem supplies local coordinates (R, y, u) with $u = \tau_{e,R}$, by Lemma 3.2. The corresponding integral is

$$\int (T - u)_+ a(R, y, u) \, dy \, du, \quad (4.1)$$

where a is smooth and compactly supported in a common chart. Differentiation in R acts only on a . Every fixed finite derivative is integrably bounded uniformly on a compact neighborhood of I . A finite covering proves $J \in C^\infty$ with the asserted finite-order bounds. The argument includes derivatives of the partition functions; in their sum they cancel by the partition identity. No analytic or order-uniform factorial bound is asserted.

4.2. Explicit derivatives. The first derivative has no level term because the factor $T - \tau$ vanishes on the moving level. Differentiating it once more gives (1.5) by the coarea formula. More explicitly,

$$\partial_R \mathbf{1}_{\{\tau_R < T\}} = -\tau'_R \delta(T - \tau_R),$$

and

$$\int f \delta(T - \tau_R) \, d\nu = \int_{\tau_R=T} \frac{f \cos \phi}{4\pi |\nabla \tau_R|} \, d\mathcal{H}^1.$$

Both identities can be verified directly in (4.1); their use does not assume a product of unrelated singular distributions.

Define the partial Bell polynomials by

$$B_{r,k}(x_1, \dots) = \sum_{\substack{\sum m_j = k \\ \sum j m_j = r}} \frac{r!}{\prod_j m_j!} \prod_j (x_j/j!)^{m_j}.$$

For $r \geq 1$ the full finite-order formula on the regular neighborhoods is

$$\begin{aligned} J^{(r)}(R) = \sum_e \int & \left[-\mathbf{1}_{\{\tau_{e,R} < T\}} \tau_{e,R}^{(r)} \right. \\ & \left. + \sum_{k=2}^r (-1)^k B_{r,k}(\tau'_{e,R}, \dots) \delta^{(k-2)}(T - \tau_{e,R}) \right] d\nu. \end{aligned} \quad (4.2)$$

The formula is understood with local cutoffs equal to one on the relevant support. It is independent of their choice. The superscript on δ differentiates its scalar argument. To prove (4.2), compose the distribution $(T - u)_+$ with $u = \tau_R$ in the flattened charts. Its first derivative in u is $-\mathbf{1}_{u < T}$ and its k th derivative is $(-1)^k \delta^{(k-2)}(T - u)$ for $k \geq 2$. The finite chain rule gives the formula. Alternatively mollify the scalar positive part, apply the ordinary chain rule, and pass to distributions using the regular-level coordinates. This proves every prescribed finite order.

The roof jets themselves have an explicit algebraic recursion. Put $h_R = \sqrt{R^2 - b_R^2} > 0$ on a short-flight branch. Then

$$h_R^{(j)} = \frac{(R^2 - b_R^2)^{(j)} - \sum_{k=1}^{j-1} \binom{j}{k} h_R^{(k)} h_R^{(j-k)}}{2h_R}, \quad \tau_R^{(j)} = L_R^{(j)} - h_R^{(j)}.$$

Here L_R and b_R are affine in R , and $h_R = Rc_1$ has a positive uniform lower bound by (3.1). Coarea derivatives are computed from the smooth density in (4.1). Finally

$$\lambda_R^{(j)} = \frac{(2R)^{(j)} - \sum_{k=1}^{\min(j,2)} \binom{j}{k} A_R^{(k)} \lambda_R^{(j-k)}}{A_R}$$

and the ordinary Leibniz rule in (1.4) determine every count-probability jet. This specifies the operations and all denominators.

5. POSITIVITY, STRICT RESPONSE, AND STATISTICAL CONSEQUENCES

5.1. A radius-uniform two-collision witness. Consider the outgoing section square $|\alpha| \leq 10^{-3}$, $|\vartheta| \leq 10^{-3}$ facing the disk at e_1 . It has $|\phi| \leq 2 \cdot 10^{-3}$. For $d = e_1 - Rn_\alpha$,

$$L = d \cdot v \leq 1 - R + 10^{-6}, \quad |b| \leq 1.94 \cdot 10^{-3}.$$

These bounds follow from $|\sin x| \leq |x|$ and $1 - \cos x \leq x^2/2$: for example $L = \cos \vartheta - R \cos(\alpha - \vartheta)$, and $|b| \leq |\sin \vartheta| + R|\sin(\alpha - \vartheta)| \leq 1.94 \cdot 10^{-3}$. Consequently,

$$0 \leq R - \sqrt{R^2 - b^2} = \frac{b^2}{R + \sqrt{R^2 - b^2}} < 5 \cdot 10^{-6}.$$

Using $L \leq 1 - R + 10^{-6}$ yields $\tau_{e_1,R} < 1 - 2R + 6 \cdot 10^{-6} < 0.10001$. The root is outgoing from the initial disk, incoming at the target, and physical by Lemma 3.1's no-obstruction argument. Hence $T - \tau > 9/1000$ throughout this square. Its ν mass exceeds $1/4,000,000$, since its area is $4 \cdot 10^{-6}$ and $\cos \phi \geq 1 - 2 \cdot 10^{-6}$. Also $\lambda_R > 9/10$ because $A_R < 1$. Therefore

$$p_2 = \lambda_R J(R) > \frac{9}{10} \frac{9}{1000} \frac{1}{4,000,000} > 10^{-9}.$$

Since $\tau'_R < 0$ throughout every active short-flight branch, (1.5) and this positive-mass square give $J'(R) > 0$. Furthermore

$$\lambda'_R = \frac{2(A_\Lambda + \pi R^2)}{A_R^2} > 0.$$

Thus $p'_2 = \lambda'_R J + \lambda_R J' > 0$.

For p_1 , use $J \leq T - d_*$ to obtain

$$p_1 \geq \lambda_R(2d_* - T) > 0.$$

For p_0 , $p_0 \geq 1 - \lambda_R T$. A coarse rational bound suffices: $A_\Lambda > 43/50$, $\pi < 22/7$, and $R \leq 47/100$ imply $A_R > 0.16$, while $2RT < 0.104$. Consequently $\lambda_R T < 0.65 < 1$. This completes the proof of Theorem 1.1.

5.2. Count generating functions and identification.

Corollary 5.1. *The count partition function is*

$$Z_R(q) = p_0 + p_1 e^q + p_2 e^{2q} = 1 + \lambda_R T(e^q - 1) + \lambda_R J(R)(e^q - 1)^2.$$

It is smooth in R , analytic in real q , and positive. Its log-source Hessian is strictly positive. Each target mean in $(0, 2)$ has a unique finite conjugate source. The binary observation $B = \mathbf{1}_{\{N_T=2\}}$ identifies R and has Fisher information

$$\mathcal{I}_B(R) = \frac{[p'_2(R)]^2}{p_2(R)(1 - p_2(R))} > 0.$$

Its square-root probability vector is differentiable in quadratic mean.

Proof. Insert (1.4) and collect powers of $e^q - 1$. All probabilities are positive, so the tilted count distribution has nonzero variance. Its mean tends to zero and two as q tends to opposite infinities, and is strictly increasing. The implicit function theorem gives local parameter derivatives of the selected source. Strict monotonicity of p_2 gives identification. Differentiating the two Bernoulli probabilities yields the information formula. Smoothness and positivity on a compact radius neighborhood give bounded second derivatives of their square roots, proving the quadratic-mean assertion by Taylor's formula. These are finite-experiment consequences, not a claim of a new general statistical principle. $\square$

6. WHAT THE NEXT MECHANICAL ESTIMATE MUST ADDRESS

The second-collision probability is obtained on the full natural preparation and includes a nonzero terminal-level contribution to its second derivative. The method does not construct the full joint two-impact path-law response: earlier and later impact directions are correlated, and sampling velocities across event times needs its own decoder analysis. It also does not supply a derivative bound on $S_j \tau_R$ uniform in j in (2.3). Those are distinct next estimates.

For A2, a concrete continuation is a source-dependent two-impact record assembly with fixed physical sampling tests, followed by a uniform bound on the complete finite-itinerary response. Long-time pressure or covariance response must use a genuinely summable version of that bound. Existing functional-analytic perturbation theory for the Lorentz gas [2] provides relevant comparison, not an automatic all-order theorem for these moving tests. The present result is an incremental mechanical input, not a replacement for that long-time task.

REFERENCES

- [1] J. Marklof, *Entry and return times for semi-flows*, arXiv:1605.02715 (2016).
<https://arxiv.org/abs/1605.02715>
- [2] M. F. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, Comm. Math. Phys. (2013); arXiv:1210.1261.
<https://arxiv.org/abs/1210.1261>
- [3] Q. Qi, *Response and fluctuations of moving-cut hyperbolic path ensembles*, manuscript of September 6, 2026, Appendix C. Source snapshot and SHA-256 are recorded in the accompanying audit. This note rederives its flux input and extends the count window; it does not import its long-time theorems into billiards.